Venkata Lakshmi Parimala Pasupuleti

Backend & AI Infrastructure Engineer · Agentic AI · PropTech · Real Estate Domain · Node.js · Python · AWS · Terraform · CI/CD

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PROFESSIONAL SUMMARY

Backend and infrastructure engineer with **8 years in property management SaaS** (Entrata, 100K+ MAU multifamily platform) and hands-on experience building **agentic AI systems** at Intuit — exactly what Zuma needs. At Entrata, served as **primary technical contact for HUD and Tax Credit federal data integrations**: onboarded external partners from scoping through go-live, owned incident response, and shipped async processing (RabbitMQ, idempotency, retries) at **99.5% uptime** across thousands of tenant records. Built **Docker microservices, AWS S3 migrations with Libsodium encryption and RBAC, and optimized 200+ SQL queries (30% gain)**. At Intuit, now ships **agentic AI features** (Anthropic Claude + Python/FastAPI + Node.js) with **GitHub Actions CI/CD, Terraform IaC, and Prometheus/Grafana observability** to AWS. Leasel is an 8-agent LangGraph platform for the rental compliance market achieving **93.6% precision**. Stack: **Node.js, Python, AWS, Terraform, Docker, Kubernetes, React, TypeScript, RabbitMQ, PostgreSQL**. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Backend & Infrastructure: Node.js, Python (9+ yrs), FastAPI, Flask, Java, Spring Boot, REST APIs, GraphQL, Async Processing, Pub/Sub (RabbitMQ, Kafka), Idempotency & Retry Logic, Microservices, Celery

Cloud & DevOps: AWS (S3, EC2, RDS, Lambda, SQS, Glue, API Gateway, IAM), GCP/Vertex AI, Terraform, Docker, Kubernetes, GitHub Actions CI/CD, Prometheus, Grafana, OpenTelemetry, Incident Response, Observability, Linux

AI & Agentic Systems: Agentic AI Workflows, LangGraph, LangChain, Anthropic Claude, OpenAI, RAG Pipelines, LLM-as-Judge Evaluation, Guardrails, Structured Outputs, ChromaDB, Pinecone, Hybrid Search

Data & Frontend: PostgreSQL, MySQL, Snowflake, Redis, MongoDB, SQLite, SQL Optimization, ETL Pipelines, React, TypeScript, Redux, HIPAA/PHI RBAC, Data Governance

WORK EXPERIENCE

Intuit

Jun 2026 – Present

Software Engineer — Agentic AI & Backend Infrastructure · Financial Analytics

- Ship **agentic AI features** integrating Anthropic Claude into Python/FastAPI + Node.js services: financial data summaries, natural-language queries, guardrails (input validation, sensitive-data handling, graceful fallbacks); deploy via **GitHub Actions CI/CD to AWS** with Terraform IaC for consistent, auditable infrastructure; **Prometheus + Grafana observability** across all LLM calls and tool invocations.
- Build and run **LLM-as-judge eval harnesses** (golden datasets + CI regression) catching hallucination and quality drift before every release; build end-to-end ML pipelines (scikit-learn, XGBoost) on Snowflake from data ingestion through prediction APIs; manage **incident response** and structured audit trails for production AI systems.

Entrata — Multifamily & Affordable Housing SaaS

Jan 2017 – Jul 2024

Senior Software Engineer — Backend Infrastructure · Async Systems · Customer Integrations

- Served as **primary technical contact for HUD and Tax Credit federal data source integrations**: scoped requirements directly with external partners, designed data contracts, coordinated go-lives, and resolved production incidents end-to-end — the exact customer onboarding and integration ownership Zuma requires. Built **RabbitMQ async processing** (pub/sub, idempotency, retry, dead-letter) for rent updates across thousands of tenant records at **99.5% uptime**.
- Shipped 20+ backend features (Python, PHP, Java, JavaScript, Node.js, React, PostgreSQL) across **120+ Agile sprints**; **optimized 200+ SQL queries (30% performance gain)** across 1M+ records; **Docker microservices** redesign (50% better scalability, 30% faster deployments); **AWS S3 migration with Libsodium encryption and RBAC** for resident PHI — 100% compliance, zero incidents; reviewed 300+ PRs and mentored junior engineers.

PROJECTS

Leasel — 8-Agent Agentic AI Platform for the Rental Compliance Market

Agentic AI · PropTech · RAG

Python · LangGraph (8-agent) · FastAPI · Hybrid RAG (TF-IDF + semantic) · OCR · SQLite · CI/CD — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production **agentic AI platform** for rental lease compliance (triage → retrieval → reasoning → validation → HITL escalation) over 1,240 unstructured documents; **OCR ingestion, section-aware chunking, hybrid RAG** with semantic reranking; guardrails routing regulated decisions to humans; **LLM-as-judge CI eval harness: 93.6% precision, 87.9% F1**. Direct analog to Zuma's rental market AI platform.

AgentProbe — Open-Source Agent Runtime & LLM Eval Toolkit

AI Infrastructure · CI Harness

Python · FastAPI · LangGraph · ChromaDB · Anthropic · OpenAI · Azure OpenAI · Groq · Ollama — github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI + Node.js backend; dynamic **model routing** across 5 LLM providers; 3 LangGraph agents; **CI golden-dataset regression harness**; LLM-as-judge scoring (accuracy, hallucination, retrieval, completeness, compliance). Ships to real users, zero external dependencies. Built as **AI infrastructure for evaluating agentic AI workflows**.

AI on FHIR — Full-Stack NLP-to-API with React + TypeScript

Full-Stack · Node.js · RBAC

Python · Flask · React · TypeScript · Node.js · Docker · BERT · LLMs · REST APIs — github.com/venkatalakshmiparimala/AI_on_FHIR

- Full-stack AI app (React/TypeScript frontend + Python/Flask + Node.js backend) translating EN/ES natural-language queries into live API calls via LLMs + BERT; **HIPAA-style RBAC** with fine-grained authorization; Docker containerized; cut manual lookup time ~60%. Proof of **React/TypeScript + Node.js full-stack delivery**.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · ML pipelines, distributed systems, LLM systems, observability, data governance

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017